Wh-movement paths and oblique extraction in Mam (Mayan)

Noah Elkins
nelkins@haverford.edu

Harold Torrence
htorrence@ucla.edu

Colin Brown
colin.brown@ubc.ca

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Abstract

This paper examines the Ā-extraction of oblique arguments and adjuncts in the Mayan language Mam (San Juan Ostuncalco dialect), focusing on the distribution of the enclitic *=(y)a'* ('MVMT'), which appears on certain predicates when an oblique is extracted. This paper has two goals. First, building on recent work on the related morpheme *=wi* in Kaqchikel and K'iche' (Mendes and Ranero 2021), we show that extraction of nearly all obliques—except temporals—triggers MVMT, and that long-distance oblique extraction triggers the enclitic in both matrix and embedded clauses, provided the embedding is sufficiently large. When a directional auxiliary is present, MVMT may appear twice in a clause: once on the verb and once on the directional. Second, we argue that *=(y)a'* in SJO is best analyzed as the spellout of Ā-agreement between the extracted oblique and specific heads along the clausal spine. We therefore propose that the enclitic diagnoses successive-cyclic movement in both monoclausal and cross-clausal *wh*-movement, providing evidence for intermediate movement through the verbal domain.

Keywords: *wh*-agreement, Ā-movement, obliques, cyclicity, Mayan, Mam

1 Introduction

This paper examines the extraction of oblique arguments and adjuncts (henceforth *obliques*) in San Juan Ostuncalco Mam (henceforth SJO), a Mamean-branch Mayan language spoken in Guatemala (iso code: `mam`). In this language, a special *movement enclitic =(y)a'* (glossed 'MVMT') surfaces on the predicate when certain obliques are Ā-extracted to the clausal left periphery. We see an example in (1). In (1a) the oblique phrase *tu'n xb'uy* 'with a rag' appears in its base-generated position. In (1b), the oblique *wh*-word *alqu'n* 'with what' is Ā-moved to clause-initial position, and the movement enclitic *=(y)a'* (optionally) appears on the predicate.

(1) Oblique extraction in San Juan Ostuncalco Mam

a. *El tsu'n Li'y spik'b'il tu'n xb'uy.*

∅-∅-el	t-su-'n	Li'y	spik'b'il	[$_{\text{OBL}}$ t-u'n	xb'uy]
COMPL-B2/3SG-DIR	A2/3SG-clean-DS	María	window	A2/3SG-RN:INS	rag

'María cleaned the window with a rag'

b. *Alqu'n el tsu'n(a') Li'y spik'b'il?*

al-qu'n$_i$	∅-∅-el	t-su-'n**(=a')**	Li'y	spik'b'il	___$_i$
what-RN:INS	COMPL-B2/3SG-DIR	A2/3SG-clean-DS=MVMT	María	window	

'With what did María clean the window?'

This paper has two main goals. The first is to describe oblique extraction in SJO, paying particular attention to which obliques trigger the appearance of the movement enclitic when extracted. We take our cue from Mendes and Ranero (2021) (henceforth M&R), who describe and analyze a similar movement particle, *wi*, in K'iche' and Kaqchikel, two Mayan languages of the K'ichean branch. We discuss local and long-distance extraction of obliques, with direct comparison to the K'ichean facts presented in M&R, and show that the distributions of SJO *=(y)a'* and K'ichean *wi* differ in several interesting ways. Specifically, we show that if the predicate is modified by a directional auxiliary, SJO *=(y)a'* may appear multiple times within a single clause (once on the verb stem and once on the directional), whereas multiple instances of *wi* within a single clause is not possible in K'ichean. Additionally, we show that in contexts of long-distance oblique extraction, *=(y)a'* occurs in both the matrix and the embedded clause, so long as the embedding is sufficiently large; in certain structurally reduced clauses, it is only possible on the matrix predicate. For K'ichean, M&R find that in contexts of long-distance oblique extraction from a sufficiently reduced embedding, *wi* is only possible in the embedding, the reverse case to SJO.

The second goal of this paper is to provide an analysis of the movement enclitic in SJO. We consider M&R's analysis, in which the movement enclitic is identified as a pronounced trace of the extracted oblique (an instance of *Chain Reduction via Substitution*), but ultimately show that it is difficult to extend this approach to SJO. We take this to suggest that the appearance of morphology that tracks oblique extraction in the two languages (K'ichean *wi* and SJO Mam *=(y)a'*) result from distinct mechanisms. In light of novel data from SJO, we propose that the movement enclitic *=(y)a'* is the spellout of an Agree relation between the relevant oblique and an Ā-head in the verbal domain. If, as we argue, multiple Ā-heads of this sort are present within a clause, we can account for the multiple clause-internal spellout of *=(y)a'* in the language. The central claim of this paper is that, if our analysis is correct, the enclitic, as a reflex of oblique extraction, diagnoses successive cyclic movement in both monoclausal and cross-clausal *wh*-movement, providing clear evidence for intermediate movement through the verbal domain (Chomsky, 1986, 1995, 2001; Rackowski and Richards, 2005; den Dikken, 2009, 2012a,b; van Urk and Richards, 2015).

This paper is organized as follows. After briefly presenting M&R's analysis in §2, and describing the relevant facts about SJO morphosyntax in §3, we present the core data on oblique extraction in the language in §4. In §5, we outline our analytical proposal concerning *=(y)a'* in the language as the spellout of Ā-agreement. We supplement our analysis in §6 by introducing additional data from a variety of long-distance extraction contexts. §7 considers, but ultimately rejects, alternative analytical possibilities for SJO *=(y)a'*. §8 concludes.

2 Theoretical background: Mendes & Ranero (2021)

This paper builds on the results in Mendes and Ranero (2021) (M&R), who analyze what they call the the *fronting particle* (glossed "FP") in K'iche' and (the Patzún variety of) Kaqchikel, two K'ichean languages of Guatemala.[1] They observe that the fronting particle in these languages, *wi*, appears immediately following the predicate when certain obliques are Ā-extracted out of that predicate's clause; its presence is obligatory in K'iche' (2), but optional in Kaqchikel (3).

(2) K'iche' (adapted from M&R, example (2))

Jawi$_i$	x-at-b'ee	*(**wi**)	iwiir	___$_i$?
where	COMPL-B2SG-go	FP	yesterday	

'Where did you go yesterday?'

[1]The K'ichean branch and the Mamean branch together form the K'ichean-Mamean, or "Eastern" Mayan sub-branch.

(3) Patzún Kaqchikel (adapted from M&R, example (3))
Ankuchi$_i$ x-a-b'e **(wi)** ___$_i$?
where COMPL-B2SG-go FP
'Where did you go?'

The relationship between the appearance of *wi* and Ā-movement is clear, based on the fact that in neither language can *in situ* obliques occur with this particle, as exemplified by Kaqchikel in (4):

(4) Patzún Kaqchikel (adapted from M&R, example (11))
Ri a Lu' x-∅-tzopin **(*wi)** chwa jay
DEF CLF Pedro COMPL-B3SG-jump FP PREP.A3.RN house
'Pedro jumped in the garden.'

Importantly, M&R note that the extraction of neither reason nor temporal obliques trigger the fronting particle in either language: for instance, we see below that the extraction of *jampe'* 'when' in Patzún Kaqchikel, and *achike ru-ma* 'why' in K'iche', does not trigger *wi*. In other words, the extraction only of *low* obliques triggers the fronting particle.

(5) Patzún Kaqchikel (adapted from M&R, example (12c))
Jampe' x-∅-in-tej knaq'?
when COMPL-B3SG-A2SG-eat beans
'When did you eat beans?'

(6) Kaqchikel (adapted from M&R, example (14a))
Achike ru-ma x-∅-samäj **(*wi)** ri a Juan?
what A3SG-RN COMPL-B3SG-work FP DET CLF Juan
'Why did Juan work?'

Under long-distance Ā-extraction of the oblique, the fronting particle can appear multiple times, once in each clause:

(7) K'iche' (adapted from M&R, example (17))
Jawi x-∅-ki-b'iij ***(wi)** [chi k-e-'e ***(wi)**] ?
where COMPL-B3SG-A3SG-say FP COMP INC-B3PL-go FP
'Where did they say that they would go?'

(8) Patzún Kaqchikel (adapted from M&R, example (18))
Ankuchi x-∅-u-b'ij **(wi)** Maria [chi x-∅-u-tej **(wi)** knaq' Juan] ?
where COMPL-B3SG-A3SG-say FP Maria COMP COMPL-B3SG-A3SG-eat FP beans Juan
'Where did Maria say that Juan ate the beans?'

Examples (7) and (8) above involve finite embedded clauses, which M&R take to be CPs because they contain an overt complementizer. Interestingly, they show that Ā-extraction of obliques from so-called *reduced* clauses pattern differently. These clauses, which are not introduced with a complementizer, are analyzed as structurally smaller Asp(ect)Ps, which lack a CP layer. As the data below show, when the embedded clause is an AspP, the fronting particle can only occur in the embedded clause.

(9) K'iche' (adapted from M&R, example (19))

Jas	r-uuk'	k-∅-a-rayii-j	**(*wi)**	[k-∅-a-tij	***(wi)**	le	wa] ?
WH	A3SG-RN	INC-B3SG-A2SG-desire-ACT	FP	INC-B3SG-A2SG-eat	FP	DET	food

'With what do you desire to eat the food?'

(10) Patzún Kaqchikel (adapted from M&R, example (20b))

Achoj	k'in	x-∅-a-tojtob'ej	**(*wi)**	x-∅-a-löq'	**(wi)**	ri	kotz'i'j?
WH	RN	COMPL-B3SG-A2SG-try	FP	COMPL-B3SG-A2SG-buy	FP	DET	flowers

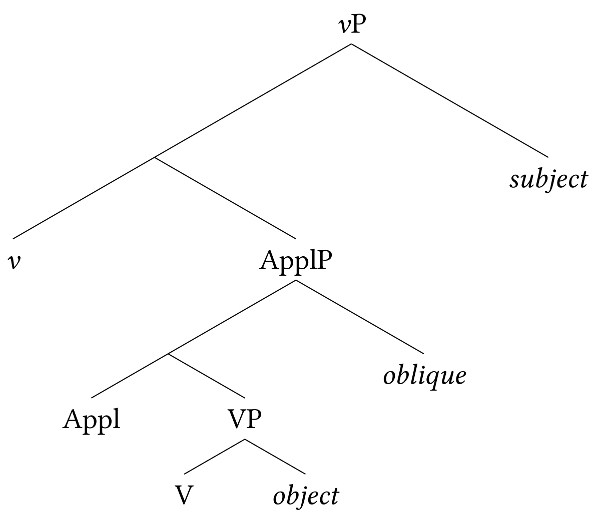
'Who did you try buying flowers with?'

Putting all of this together, M&R formulate the following generalization, which they take to be true of Kaqchikel and K'iche':

(11) **Fronting Particle Generalization:** In long distance Ā-extraction of low oblique phrases from a single embedded clause, the presence of *wi* in the matrix clause is contingent on the presence of an overt complementizer in the embedded clause. (modified from M&R (21))

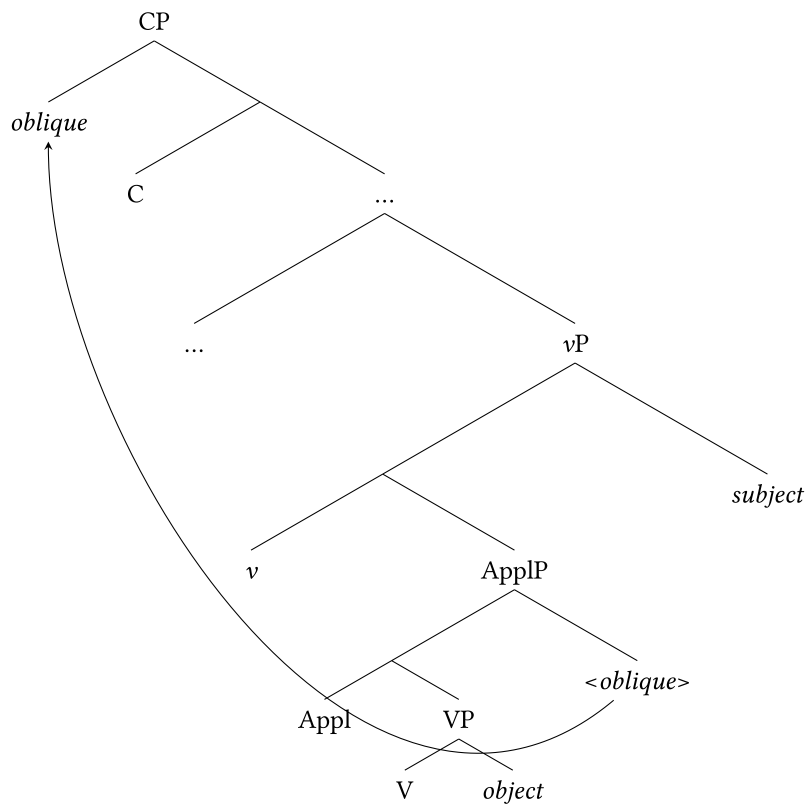
To account for the Fronting Particle Generalization, they argue that low oblique phrases are merged as specifiers of a left-branching Applicative Phrase (ApplP) that sits between v^0 and VP (they propose that Appl^0 is null). The structure they propose is given in (12):

(12)



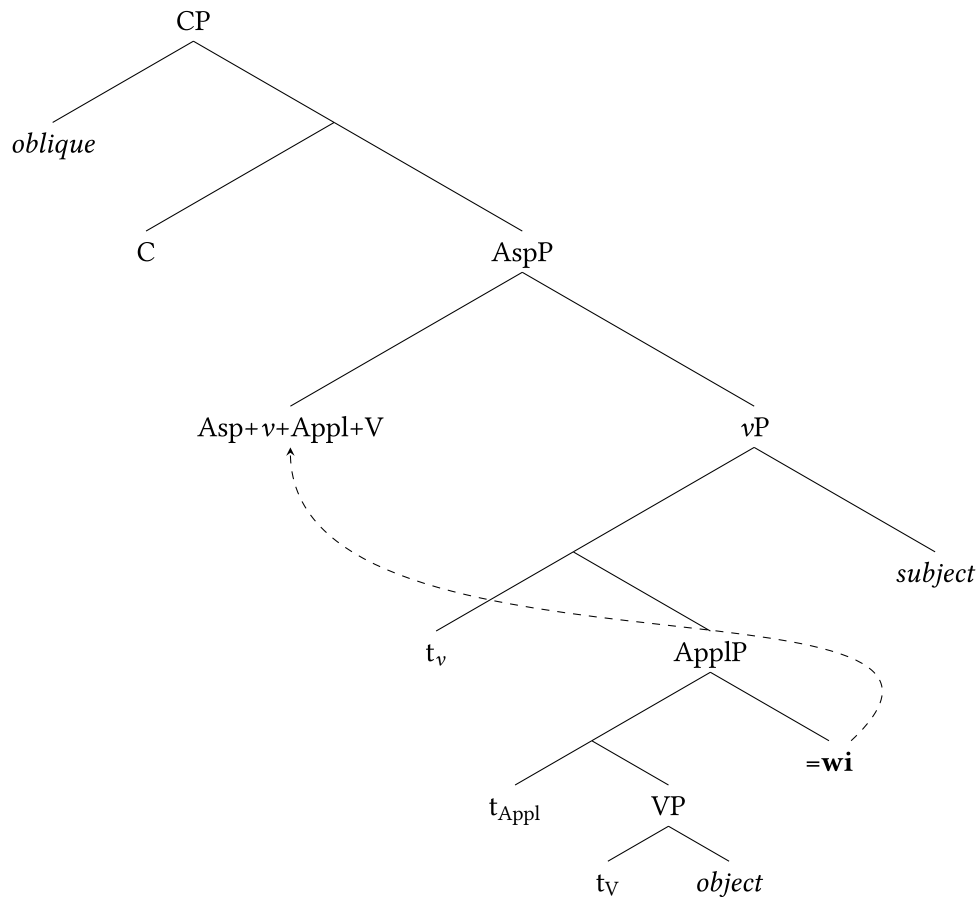
The relevant configuration occurs when the oblique Ā–moves to Spec,CP:

(13)



The central issue for M&R is what happens to the trace of the moved oblique (in Spec,ApplP). Under the Copy Theory of Movement (Chomsky 1993; Nunes 2004 and much subsequent work), movement (Internal Merge) yields multiple copies. To account for why typically only one copy is pronounced, the PF process of Chain Reduction (CR) is invoked. As it is generally understood, this operation obviates any linearization paradoxes emerging through the existence of multiple copies by deleting all but one copy (usually the lower one(s)) of the relevant syntactic object (Chain Reduction via Deletion). However, M&R propose that the grammar allows for an alternative PF operation: *Chain Reduction via Substitution*, by which the lower copy/copies are substituted by a special vocabulary item. M&R propose that the fronting particle in Kaqchikel and K'iche' is the instantiation of just such a vocabulary item: in these languages, Chain Reduction via Substitution results in the lower copies of the relevant obliques being realized as *wi* at PF. The difference between the two languages is that in K'iche', the substitution is obligatory, leading to the fronting particle *wi* also being obligatory. In Kaqchikel, on the other hand, application of the substitution rule is optional; therefore, the lower copy can be realized as either ∅ or *wi*. Thus, in a case like (7) above, Ā-movement results in two copies of the oblique: one in Spec,CP and another in Spec,ApplP, with Chain Reduction operating in both languages. The tree in (14) below, adapted from M&R (31) shows the case in K'iche', where Chain Reduction via Substitution is obligatory.

(14)



As the schematization above shows, M&R assume that *wi* postsyntactically encliticizes to the predicate (therefore, M&R analyze *wi* as a *special clitic* (Zwicky, 1977; Zwicky and Pullum, 1983): its surface position is distinct from its base-generated position). In what follows, we show, based on novel data from SJO, that this analysis does not readily extend to account for the distribution of the movement enclitic in Mam. We present our analysis in §5, and return to M&R's proposal in §8. First, we turn to a brief background on Mam.

3 Background on San Juan Ostuncalco Mam

The empirical focus of this paper is Mam, an Eastern Mayan language of the Mamean branch spoken by approximately 600,000 people in the western highlands of Guatemala. We examine the underdocumented variety spoken in San Juan Ostuncalco, a member of the Southern Mam dialect area (England, 2017). It should be noted, however, that having a movement enclitic in Mam is not particular to this dialect, nor to the Southern Mam dialect group: it is also found in the Western Mam dialect group, specifically in Tacaná Mam (Munson 1984). In this section, we lay out the relevant morphosyntactic background information that will be relevant to the discussion of the movement enclitic in SJO. All uncited Mam data from this paper were elicited from three speakers of SJO Mam between 2023-2026 either *in situ* in Quetzaltenango and San Juan Ostuncalco, Guatemala or over telecommunications platforms.

3.1 Mam morphosyntax

SJO Mam exhibits ergative-absolutive alignment, with case relations marked on the predicate (*head-marking*). In the Mayanist tradition, ergative (and possessive) person markers are referred to as 'Set A', and absolutive person markers as 'Set B'. Some persons in both Sets A and B also trigger a predicate-final enclitic. 2nd and 3rd person (singular and plural) and 1st person plural inclusive and exclusive are syncretic; these are distinguished by the enclitic. Paradigms for Sets A and B are given below.

Person	Singular	Plural
1EXCL	*n- ∼ w- ... =(y)e'*	*q- ... =(y)e'*
1INCL		*q-*
2	*t- ... =(y)a*	*k- ... =(y)e'*
3	*t-*	*k-*

Table 1: Set A person marking

Person	Singular	Plural
1EXCL	*chin ... =(y)e'*	*qo ... =(y)e'*
1INCL		*qo*
2	*tz'- ∼ ∅- ... =(y)a*	*chi ... =(y)e'*
3	*tz'- ∼ ∅-*	*chi*

Table 2: Set B person marking

Like all Mam varieties, SJO has VSO word order (England, 1991): in broad-focus transitives with two overt arguments, the subject always precedes the object (see also Elkins 2025). The verb is a complex constituent with multiple slots. In initial position in the verbal complex, aspect/mood (AM) or negation appears. Set B precedes Set A, though a directional auxiliary (Dir) may intervene. Set A prefixes attach to the verb stem, which can itself host verbal suffixes marking transitivity. Enclitics—including person markers and the movement enclitic *=(y)a'*—follow these suffixes, with *=(y)a'* always outermost. The basic clausal template is shown in (15).

(15) Mam clausal template
AM/NEG SET.B DIR $_V$[SET.A-**Stem**-suffixes]=enclitics (*subject*) (*object*)

Oblique/non-core arguments (X) appear in a clause-final position.[2] A full, finite transitive VSOX clause is given in (16).

(16) *El tsu'n Li'y spik'b'il tu'n xb'uy.*

∅-∅-el	t-su-'n	[Li'y]$_S$	[spik'b'il]$_O$	[t-u'n	xb'uy]$_X$
COMPL-B2/3SG-DIR	A2/3SG-clean-DS	María	window	A2/3SG-RN:INS	rag

'María cleaned the window with a rag.'

[2]As we will see below, oblique phrases may also appear in clause-initial position if they are focused or *wh*-questioned.

Oblique arguments in Mam are typically introduced by means of relational nouns (RNs), such as *u'n* in (16) above. These have the function of prepositions in other languages, but are notable for (i) being derived from body part nominals, and (ii) agreeing with their complements with Set A agreement. In (16), the 3sg *xbu'y* 'rag' is co-indexed by the Set A prefix *t-* on the RN.

3.2 Core argument extraction in SJO

Focused, relativized, and *wh*-questioned elements are obligatorily preposed to a clause-initial position; we refer to this preposing process as *extraction.* For core arguments, the extraction of absolutives (intransitive subjects and transitive objects) is unmarked, as shown in examples (17) and (18).

(17) Absolutive extraction: intransitive subjects

a. *Kyim jun xjaal.*
∅-∅-kyim jun xjaal
compl-b2/3s-die indf man
'A man died'

b. *Alkye kyim?*
alkye ∅-∅-kyim ___
who compl-b2/3s-die
'Who died?'

(18) Absolutive extraction: transitive objects

a. *Tjaq' Li'y tpjel ja'.*
∅-∅-t-jaq' Li'y tpjel ja'
compl-b2/3sg-a2/3sg-open María door house
'María opened the door.'

b. *Titi' tjaq' Li'y?*
titi' ∅-∅-t-jaq' Li'y ___
what compl-b2/3sg-a2/3sg-open María
'What did María open?'

The extraction of ergatives (transitive subjects) is not straightforwardly possible, due to Mam's sensitivity to the Ergative Extraction Constraint (EEC), which is common among Mayan languages (see Coon and Preminger 2012; Coon et al. 2021; Aissen 2017 and references therein). In SJO, an antipassive construction is required to extract ergative arguments. The antipassive theme argument is introduced by a relational noun, which features Set A marking indexing agreement with the φ-features of that argument. The antipassive verb is additionally marked with an intransitive/antipassive suffix *-n* 'antip' (19).

(19) Ergative extraction triggers antipassive

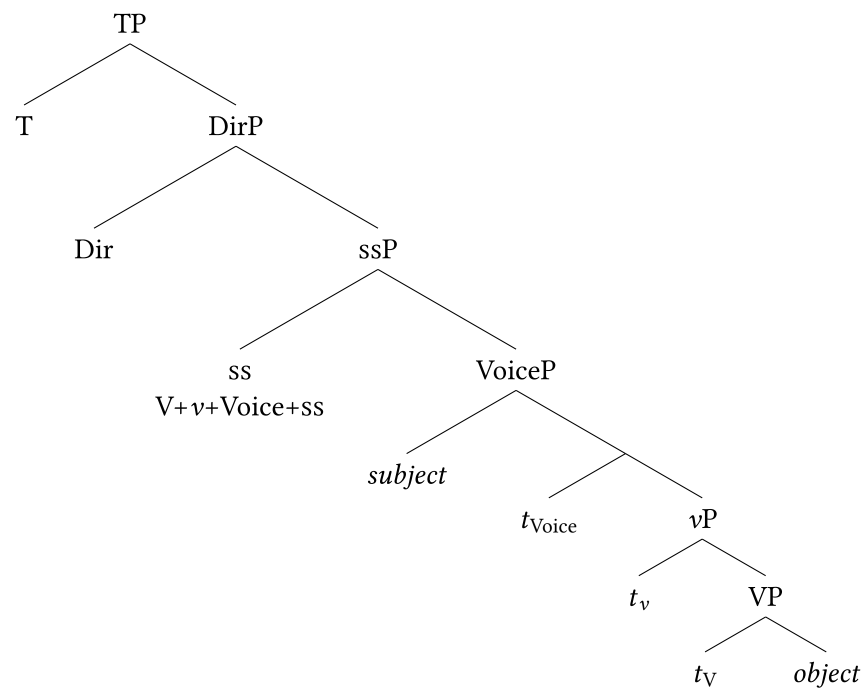
a. *Jaw tjq'o'n Li'y tjpel ja'.*
∅-∅-jaw t-jq'o-'n Li'y tjpel ja'
compl-b2/3sg-dir a2/3sg-open-ds María door house
'María opened the door.'

b. *Alkye jq'on te tjpel ja'?*
alkye ∅-∅-jq'o-**n** ___ [OBL **t-e** tjpel ja']
who COMPL-B2/3S-open-ANTIP A2/3S-RN:PAT door house
'Who opened the door?'

Importantly for the following discussion, the extraction of core arguments does *not* involve the use of the movement enclitic *=(y)a'*. As we will see in more detail in §4, this enclitic only appears when an oblique argument is extracted to clause-initial position.

3.3 Structural assumptions

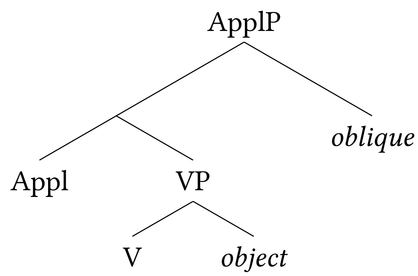
Before turning to the core empirical data on oblique extraction in SJO Mam, we outline the theoretical assumptions concerning clausal syntax adopted in this paper. While the precise hierarchy of functional structure in Mam remains an open issue, we follow recent work on Mam clausal syntax (Scott 2023; Elkins 2023, 2025; Elkins et al. 2025), in assuming that verb-initiality in the language is achieved by means of roll-up head movement of the verb through to a projection labelled ssP (for 'status suffix phrase', after Clemens and Coon 2018). The fact that a directional precede the verb stem in Mam is accounted for by means of a high DirP merging above ssP, the head of which realizes the directional auxiliary. Transitive subjects are introduced in the specifier of VoiceP (Burukina and Polinsky 2025). The derivation of predicate-initial word order is schematized below in (20).[3]

(20)


[3]Not shown here is an additional operation of *object raising*, independently needed to account for syntactic ergativity/EEC-sensitivity in high-ABS Mayan languages like Mam (e.g. Coon et al., 2014; Royer, 2022; Royer and Coon, 2025; Myers, 2025). Because it does not influence the present discussion, we remain agnostic as to whether this object raising is overt and rightward, under a right-oriented specifiers approach (Aissen, 1992; Little, 2020; Scott, 2023; Elkins, 2025), or whether it is covert and leftward (Coon et al., 2014).

Additionally, we follow Mendes and Ranero (2021) in assuming that the relevant obliques — those that trigger the movement enclitic when extracted — form a natural class in that they are introduced in the specifier of Appl(icative)P (Pylkkänen 2002, 2008), whose head is phonologically null; this projection is a 'high applicative' in the terminology of Pylkkänen.[4]

(21)



4 Oblique extraction

In this section, we turn to the distribution of the movement enclitic *=(y)a'*, in monoclausal oblique extraction configurations. Specifically, we examine the extraction of instruments (§4.1), benefactives and datives (§4.2), locatives (§4.3), reasons and purposes (§4.4), manners (§4.5), and temporals (§4.6), and show that in all cases except for temporal extraction, the movement enclitic is licensed. We additionally show that the movement enclitic can surface multiple times in a single clause (§4.7). The descriptive generalizations presented here are are analyzed in §5. Finally, we note that while our discussion focuses on *wh*-movement, both focus movement and relativization of obliques exhibits the same distributional pattern with respect to the presence (or absence) of *=(y)a'*.

4.1 Instruments

Instruments are introduced with the relational noun *(q)u'n* 'RN:INS' (22a).[5] In (22b), we see that the *wh*-extraction of this instrument oblique optionally triggers the movement enclitic *=(y)a'*.[6]

(22) Instrument extraction

a. *El tsu'n(*a') Li'y spik'b'il tu'n xb'uy.*

∅-∅-el	t-su-'n(***=a'**)	Li'y	spik'b'il	[t-u'n	xb'uy]
COMPL-B2/3SG-DIR	A2/3SG-clean-DS=MVMT	María	window	A2/3SG-RN:INS	rag

'María cleaned the window with a rag'

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[4]In (21), the same structural properties could also be derived with a leftward specifier plus independently motivated subject and object movement, yielding VSOX order. Since the choice between these implementations does not affect the core analysis, we do not attempt to distinguish between them here.

[5]Relational nouns precede their nominal complements except when appearing alongside *wh*-words in *wh*-interrogatives (compare the order *t-u'n xb'uy* 'with a rag' in (22a) and *al-qu'n* 'with what' in (22b)). This is a case of so-called *pied piping with inversion* (Smith Stark, 1988) or *secondary wh-movement* (Heck, 2008). We do not pursue an analysis of this phenomenon in this paper, but see Heck (2008); Coon (2009); Aissen and Polian (2025); Royer et al. (2025). Additionally, we note here that the *q*-initial allomorph of relational nouns is restricted to these pied-piping with inversion contexts.

[6]Whether the glide <y> is present or not is entirely phonological: it appears if the predicate-final segment is a vowel or glottal stop <'>.

b. *Alqu'n el tsu'n(a') Li'y spik'b'il?*

al-qu'n	∅-∅-el	t-su-'n**(=a')**	Li'y	spik'b'il ___
what-RN:INS	COMPL-B2/3SG-DIR	A2/3SG-clean-DS=MVMT	María	window

'With what did María clean the window?'

We additionally see that in (22a), when the adjunct is in-situ, the movement enclitic is impossible. Thus, the movement enclitic is only associated with extraction of the instrument, not merely its presence.

4.2 Benefactives and datives

Benefactives and datives are both introduced by the relational noun *(q)e*; both are *wh*-questioned as *alqe*. In (23), we see that the extraction of *(q)e* in its benefactive reading optionally triggers the movement enclitic *=(y)a'*, as does the extraction of *(q)e* in its dative reading (24). Just as with instruments, example (23a) shows that in-situ obliques cannot trigger the movement enclitic.

(23) Benefactive extraction

a. *Jaw tjq'o'n(*-a') Li'y tjpel ja' te Pedro.*

∅-∅-jaw	t-jq'o-'n**(*=a')**	Li'y	tjpel	ja'	[t-e	Pedro]
COMPL-B2/3SG-DIR	A2/3SG-open-DS=MVMT	María	door	house	A2/3SG-RN:BEN	Pedro

'María opened the door for Pedro'

b. *Alqe tjaq'(a')ya tjpel ja'?*

al-qe	∅-∅-t-jaq'**(=a')**=ya	tjpel	ja' ___
who-RN:BEN	COMPL-B2/3SG-A2/3SG-open=MVMT=2P	door	house

'For whom did you open the door?'

(24) Dative extraction

a. *Xi' kq'o'ne' saqchb'il kye k'wal.*

∅-∅-xi'	k-q'o-'n=e'	saqchb'il	[ky-e	k'wal]
COMPL-B2/3SG-DIR	A2/3SG-give-DS=2P	toy	A2/3P-RN:DAT	child

'Y'all gave the toys to the children'

b. *Alqe' xi' kq'o'n(a)ye' saqchb'il?*

al-qe	∅-∅-xi'	k-q'o-'n**(=a')**=ye'	ksaqchb'il ___
who-RN:DAT	COMPL-B2/3SG-DIR	A2/3P-give-DS=MVMT=2P	children

'To whom did you give the toys?'

4.3 Locatives

The extraction of a locative oblique also allows the use of the movement enclitic *=(y)a.* In (25a), the locative expression is *toj b'e'* 'in the road' follows the VSO string and the movement enclitic is ungrammatical. However, in (25b), we see that *=(y)a'* may appear on the predicate in locative questions featuring the *wh*-word *ja* 'where?'.

(25) Locative extraction[7]

a. *E' chiyon tx'yan toj b'e'.*
∅-e' chiyo-n(***=a'**) tx'yan [t-oj b'e']
COMPL-B2/3S bark-ANTIP=MVMT dog A2/3S-RN:in road
'The dog barked in the road.'

b. *Ja e' chiyon(a') tx'yan ewi?*
ja ∅-e' chiyo-n(**=a'**) tx'yan ___
where COMPL-B2/3S bark-ANTIP-MVMT dog
'Where did the dog bark?'

4.4 Reasons and purposes

Next, we examine the extraction of reason and purpose obliques, both of which in SJO involve *tiqu'n*, the *wh*-expression for 'why'. The use of *tiqu'n* optionally licenses marking on the predicate by the movement enclitic. This pattern holds both when *tiqu'n* receives a reason interpretation (as in (26b)) and when it receives a purpose interpretation (as in (27b)).

(26) Reason extraction

a. *El tsu'n Li'y spik'b'il tu'n xb'uy tu'n k'ejil*
∅-∅-el t-su-'n Li'y spik'b'il t-u'n xb'uy [t-u'n k'ejil]
COMPL-B2/3SG-DIR A2/3SG-clean-DS María window A2/3SG-RN:INS rag A2/3S-RN:REASON lazy
'María cleaned the window with a rag because she is lazy'

b. *Tiqu'n el tsu'n(a') Li'y spik'b'il tu'n xb'uy?*
tiqu'n ∅-∅-el t-su-'n(**=a'**) Li'y spik'b'il t-u'n xb'uy ___
why COMPL-B2/3SG-DIR A2/3SG-clean-DS=MVMT María window A2/3SG-RN:INS rag
'Why did María clean the window with a rag?'

(27) Purpose extraction

a. *Tjoy Xwan tx'yol toj k'ul tu'n kxi' tk'a'yin toj k'wik.*
∅-∅-t-joy Xwan tx'yol t-oj k'ul [t-u'n k-xi' t-k'a'y-in t-oj k'wik]
COMPL-B2/3S-A2/3S-look.for Juan mushroom A2/3S-RN:in forest A2/3S-RN:PURP A2/3P-DIR A2/3S-sell-DS A2/3S-RN:in market
'Xwan looked for mushrooms in the forest in order to sell them in the market'

b. *Tiqu'n tjoy(a') Xwan tx'yol toj k'ul?*
tiqu'n ∅-∅-t-joy(**=a'**) Xwan tx'yol t-oj k'ul ___
why COMPL-B2/3S-A2/3S-look.for=MVMT Juan mushroom A2/3S-RN:in forest
'Why did Juan look for mushrooms in the forest?'

[7] The allomorph of B2/3S seen in this example, *-e'*, is rare: it appears only following the null allomorph of the completive aspect and when there is no directional following.

4.5 Manners

The extraction of manner obliques works similarly: the *wh*-expression for 'how' in SJO is *se'n(=tu'mel)*.[8] The use of this *wh*-expression optionally triggers the use of *=(y)a'*.

(28) Manner question with *se'n=tu'mel* 'how'

a. *Kub' tpa'n Li'y lmet*

∅-∅-kub'	t-pa-'n	Li'y	lmet
COMPL-B2/3S-DIR	A2/3S-break-DS	María	bottle

'María broke the bottle'

b. *Se'ntu'mel kub' tpa'n(a') Li'y lmet?*

se'n=tu'mel	∅-∅-kub'	t-pa-'n**(=a')**	Li'y	lmet ___
how=ENCL	COMPL-B2/3SG-DIR	A2/3SG-break-DS=MVMT	María	bottle

'How did María break the bottle?'

4.6 Temporals

In contrast to the previous cases, we show that the extraction of temporal obliques in SJO Mam patterns differently from the extraction of other obliques, in that it does *not* trigger the appearance of *=(y)a'*. SJO has two temporal *wh*-expressions: *jtoj* is used only in non-past clauses (29a), and *jtoje* is used in past clauses (29b). The extraction of neither of these elements is associated with the appearance of the movement enclitic.

(29) a. *Jtoj kxe'l telq'a'n Xwan chmek'?*

jtoj/*jtoje	k-xe'-l	t-elq'a-'n**(*=a')**	Xwan	chmek'
when.NON.PST/when.PST	B2/3S-DIR-POT	A2/3S-steal-DS=**MVMT**	Juan	turkey

'When will Juan steal the turkey?'

b. *Jtoje xi' telq'a'n Xwan chmek'?*

jtoje/*jtoj	∅-∅-xi'	t-elq'a-'n**(*=a')**	Xwan	chmek'
when.PST/when.NON.PST	COMPL-B2/3S-DIR	A2/3S-steal-DS=**MVMT**	Juan	turkey

'When did Juan steal the turkey?'

Temporals appear to be the only oblique arguments which do not trigger the movement enclitic when Ā-extracted (see also §4.8). We discuss why this might specifically be the case for Mam in our discussion below (§8).

4.7 Multiple spellout of the enclitic

Here, we introduce an additional property of the spellout of the movement enclitic *=(y)a'* in SJO that distinguishes it from its K'ichean counterpart *wi*. In the preceding examples, the movement enclitic surfaced only once per clause. In this section, however, we show that *=(y)a'* may be realized multiple times within a single clause. Specifically, in addition to appearing on the verb stem, a separate instance of the movement

[8] Certain *wh*-expressions in Mam appear with optional lexically-specific emphatic enclitics; for 'how', which is *se'n*, the its enclitic is *=tumil*, glossed 'ENCL' above. This enclitic is also found optionally with 'where', which is *ja(=tu'mel)*.

enclitic can also appear on a directional auxiliary, when one is present. In (30) below, we see that in a monoclausal context where the oblique *ja* 'where' undergoes extraction, the movement enclitic appears on the predicate *xhchin* 'shouts' and additionally on the preceding directional *jaw*. Both instances are optional.

(30) Cyclic spellout of *=(y)a'*: monoclausal context (intransitive)
Ja jaw(a') xhchin(a') Luch?

ja	∅-∅-jaw**(=a')**	xhchi-n**(=a')**	Luch
where	COMPL-B2/3S-DIR=**MVMT**	shout-ANTIP=**MVMT**	Pedro

'Where did Pedro shout?'

This phenomenon occurs regardless of which lexical verb, which directional, or which oblique expression is present in the relevant clause. There is additionally no difference between intransitive clauses, like the one above, and transitive clauses: see, for instance, the following example. We discuss these facts further in §8.

(31) Cyclic spellout of *=(y)a'*: monoclausal context (transitive)
Se'ntumel e' xi'(ya') telq'an(a') Li'y u'j?

se'n=tumel	e'	∅-xi'**(=ya')**	t-elq'a-'n**(=a')**	Li'y	u'j?
how=ENCL	COMPL	B2/3S-DIR=**MVMT**	A2/3S-rob-DS=**MVMT**	María	book

'How did María steal the book?'

4.8 Local summary and comparison with K'ichean

To summarize the preceding data, in SJO Mam the extraction of the following obliques optionally triggers the movement enclitic *=(y)a'* on the clauses' predicate: instruments (§4.1), benefactives/datives (§4.2), locatives (§4.3), reasons/purposes (§4.4), and manners (§4.5). The extraction of temporals (§4.6) is unique in that it is *not* associated with the appearance of *=(y)a'*. At this point, we can compare the contexts that condition the appearance of the movement enclitic in Mam with those that do so in the K'ichean languages, where the corresponding element is *wi* (M&R).

In K'ichean, only a subset of oblique arguments — specifically the so-called *low* obliques — trigger the appearance of *wi* under extraction, while clausal or *high* obliques, such as reason and temporal adverbs, do not. (Relevant examples are given above; compare (2)–(3) with (5)–(6)). The contexts that give rise to the movement enclitic in Mam and K'ichean are largely, though not entirely, overlapping. Table 3 summarizes the environments in which the relevant movement enclitic appears in both Mam and K'ichean.[9]

[9]The extraction of manner obliques was not tested by M&R, but one author (Rodrigo Ranero, p.c.) reports that *wi* is not possible in this context in Kaqchikel; it is likewise impossible in K'iche', as confirmed by Sapón and Can Pixabaj (2000):204–205.

Oblique Type	SJO Mam	K'ichean
Instruments	✓	✓
Benefactives/datives	✓	✓
Locatives	✓	✓
Temporals	✗	✗
Reasons/purposes	✓	✗
Manners	✓	✗

Table 3: Comparison of oblique extraction across SJO Mam and K'ichean

Among those contexts that are investigated both here for Mam and by M&R for K'ichean, there is a mismatch with the behavior of reason and manner oblique extraction (highlighted in the table above): in Mam, the extraction of reasons like *tiqu'n* 'why' and *se'n(=tu'mel)* 'how' require the movement particle *=(y)a'*, whereas 'why' and 'how' in K'ichean obligatorily lack the movement particle *wi* when extracted (compare, e.g., Patzún K'iche' (6) with Mam (26b)). M&R propose for K'ichean that these high obliques are base-generated in a position higher than Spec,ApplP, unlike the rest of the (low) obliques; for this reason, the insertion of the vocabulary item *wi*, which is sensitive to the presence of an [APPL] feature, does not apply. Given this analysis, the present findings are unexpected: why is it that 'how' and 'why' — which are, by hypothesis, high obliques — still trigger *=(y)a'* in SJO when extracted? A preliminary suggestion would be that the subset of obliques counting as either 'high' or 'low' is subject to variation in different languages. In K'ichean, for instance, 'why?'-questions involve the base-generation of a high oblique, whereas in SJO, the equivalent oblique is base-generated low (i.e. in Spec,ApplP). It should be noted that, even among K'ichean, there is notable variation. In one dialect of K'iche, for instance, López Ixcoy (1997):410-413 notes that the fronting particle *wi* is required for the extraction of an indirect object (dative), but not for benefactives. In another dialect of the same language, Sapón and Can Pixabaj (2000):204-205 report that *wi* is required for benefactives. A thorough investigation of which obliques are to be classed as 'high' or 'low' in K'ichean (and Mam) dialects is beyond the scope of this paper, but we believe future investigation will be able to shed light on this question. We do, however, further discuss the special case of temporal obliques in Mam in Section 8 below.

5 Analysis

In this section, we argue that the spellout of the movement enclitic *=(y)a'* in SJO Mam is a reflex of clause-internal cyclic movement (Chomsky, 1986, 1995, 2001; Rackowski and Richards, 2005; den Dikken, 2009, 2012a,b; van Urk and Richards, 2015). We analyze *=(y)a'* as the morphological reflex of Ā-agreement (e.g., Chung and Georgopoulos, 1988; Georgopoulos, 1991; Ouhalla, 1993; Chung, 1994; Baier, 2018) between a moved oblique and successive heads along the clausal spine, specifically Voice^0 and Dir^0. On this view, each occurrence of *=(y)a'* corresponds to a distinct step in the movement path. This analysis therefore derives the otherwise unexpected pattern discussed in (§4.7): the enclitic may surface multiple times within a single clause precisely because the oblique argument undergoes (clause-internal) successive-cyclic movement.

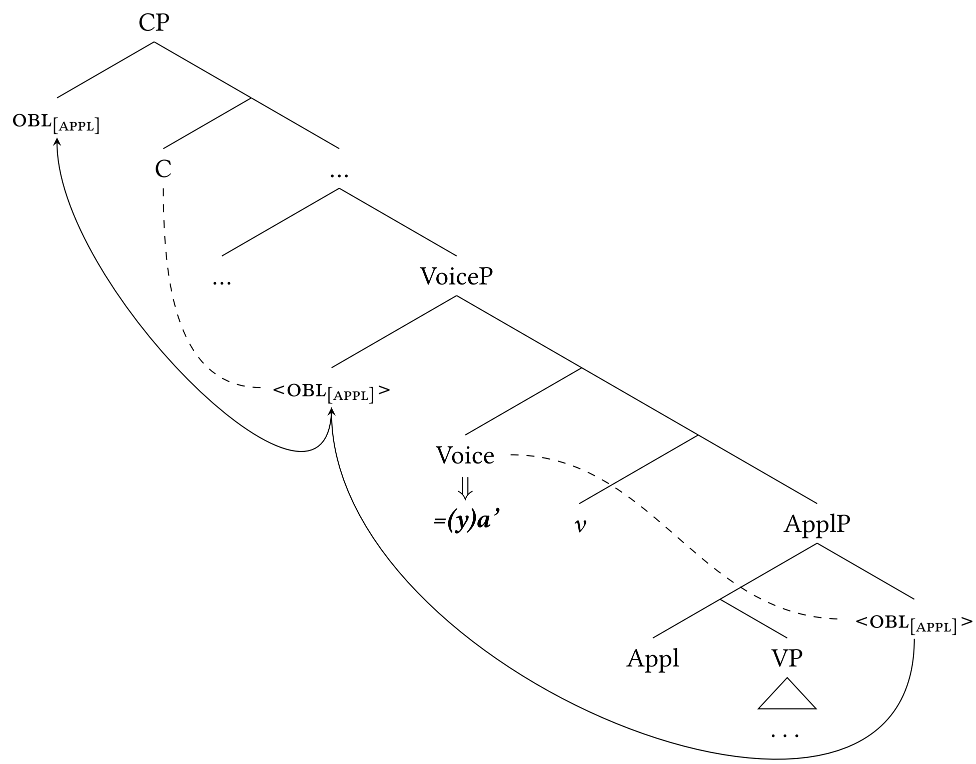
We extend this analysis to long-distance extraction in Section 6, and address possible alternative analyses in §7.

5.1 The movement enclitic as the spellout of Ā-agreement

Our analysis of oblique extraction and the spellout of *=(y)a'* diverges from that which M&R proposes for K'ichean *wi*. Instead, we propose that *=(y)a'* is the spellout of an Agree relation between the relevant obliques and an Ā-head in the verbal domain. When any element is Ā-extracted, it passes through the specifier position of this Ā-head. If the element is a core argument, the spellout of that agreement is null; if the element is an oblique, the vocabulary item *=(y)a'* is generated at that position.

Let us begin with a derivation of oblique extraction within a single clause. As outlined in our theoretical assumptions above (§3.3), we follow M&R in assuming that the relevant obliques are base-generated in Spec,ApplP and receive an [APPL] feature via a Spec–head relationship with Appl^0. In extraction contexts, these obliques additionally bear an [Ā] feature. After ApplP and *v*P are composed, Voice^0 is merged. We propose that this head bears an Ā-probe that searches its c-command domain, Agrees with the oblique, and raises it to Spec,VoiceP. The morphological reflex of this Agree relation between the oblique and Voice^0 is realized as *=(y)a'*. The oblique subsequently undergoes further Agree with C^0, triggering movement to Spec,CP. The derivation for the monoclausal sentence in (25b) is given in (32), where the dashed lines indicate agreement and solid lines indicate movement. The movement enclitic is encliticized to the verb stem following head movement to this position (not shown).

(32)



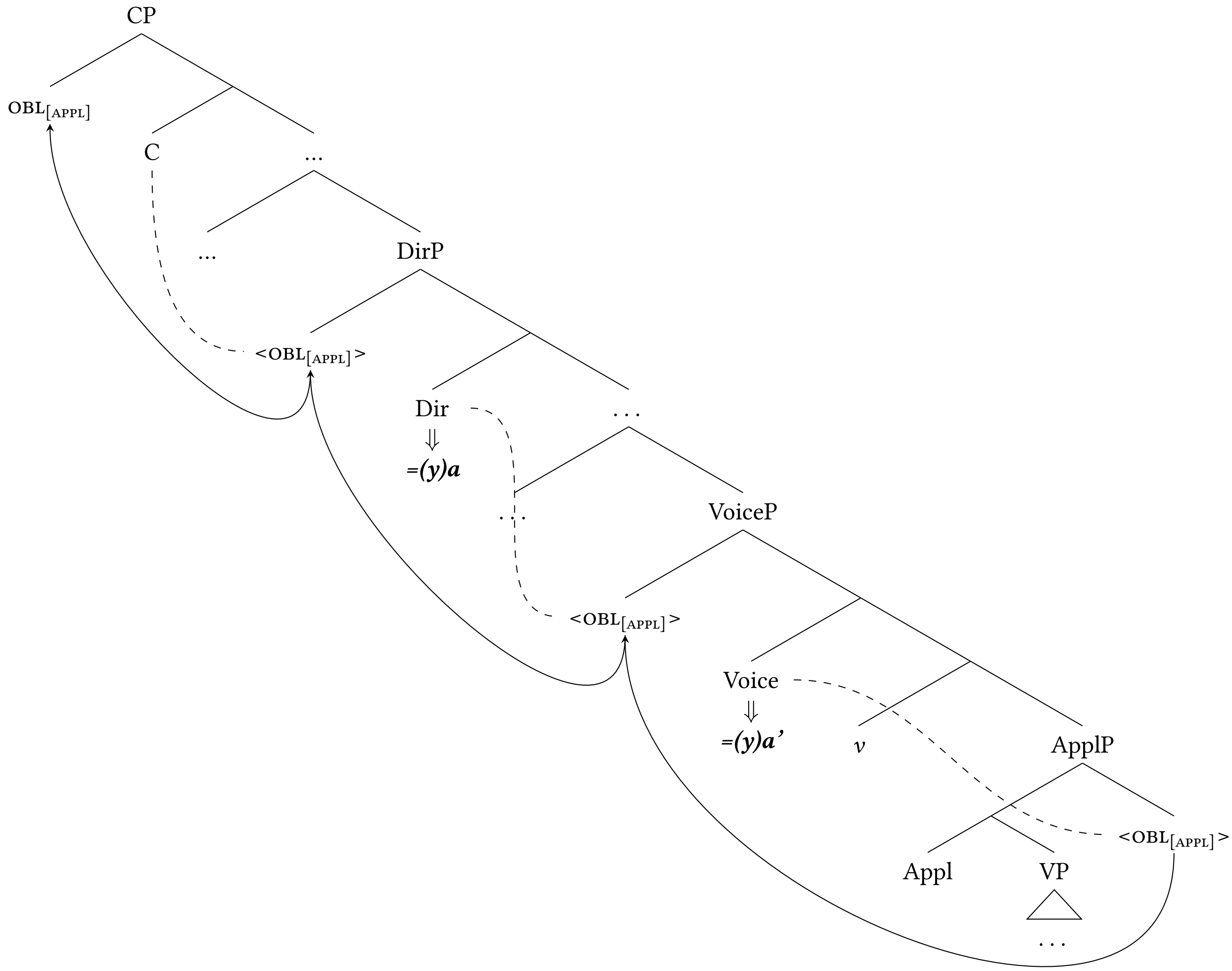
The analysis of *=(y)a'* as the spellout of Ā-agreement helps us account for multiple spellout as well. All that we need additionally assume is that the head of Dir(ectional)P also bears an Ā-probe analogous to the one on Voice. In such cases, instead of undergoing just one intermediate step of movement before reaching Spec,CP (as in 32), there will be two intermediate steps of movement, one to Spec,VoiceP and then to Spec,DirP: both intermediate steps will trigger the spellout of *=(y)a'* in such cases.

(33) *Se'ntumel e' xi'(ya') telq'an(a') Li'y u'j?*

se'n=tumel	e'	∅-xi'**(=ya')**	t-elq'a-'n**(=a')**	Li'y	u'j
how=ENCL	COMPL	B2/3S-DIR=**MVMT**	A2/3S-rob-DS=**MVMT**	María	book

'How did María steal the book?'

(34) Multiple spellout of *=ya'*



As we stated in §3.3 regarding our structural assumptions, the main verb and the directional auxiliary (if present) are within a single clause: the directional+verb complex is not an instance of a serial verb construction or clause chaining. We therefore take examples like (30) and (31) to be genuine cases of multiple spellout *within a single clause*. As a reflex of the extracted oblique transiting multiple specifier (gap) positions within the middlefield of the clause, therefore, multiple spellout of *=(y)a'* in SJO provides clear evidence for clause-internal successive cyclicity.

In the next section, we introduce long-distance oblique extraction in Mam, and show that the same movement enclitic appears in each clause that the movement proceeds through. We extend the analysis presented here to these long-distance configurations in Section 6.2.

6 Long-distance oblique extraction

Having presented our analysis of the movement enclitic *=(y)a'* in monoclausal extraction configurations, we now provide additional data in favor of our approach from the domain of long-distance (cross-clausal) oblique extraction, and extend our analysis to these cross-clausal contexts.

6.1 Extraction from CP-sized and reduced clauses in K'ichean

Before we turn to the behavior of long-distance oblique extraction in Mam, we outline M&R's analysis of long-distance oblique extraction in K'ichean.

In their survey of the movement particle *wi* in K'ichean, M&R find that the extraction of the relevant oblique phrases may lead to the appearance of multiple instances of *wi*, just in case the lower clause is a full CP. In such contexts, both the matrix and embedded predicates are encliticized with *wi* (35); in this example, full CP status of the embedded clause is indicated by the presence of the complementizer *chi* 'COMP', which is not present in structurally smaller embedded clause types.

(35) K'iche' extraction from CP: multiple *wi* (M&R 2021:11); embedding bracketed

Jawi	x-∅-ki-b'iij	*(**wi**)	[	chi	k-e-'e	*(**wi**)	]$_{CP}$	?
where	COM-B3S-A3P-say	MVMT		COMP	INC-B3P-go	**MVMT**		

'Where did they say that they would go?'

On the other hand, they find that extraction across the boundary of a clause smaller than a full CP does not lead to multiple appearances of *wi*. Instead, *wi* is only possible in the embedded clause: the matrix clause predicate obligatorily lacks *wi* (36); in this following example, the clause is a small Asp(ect)P, with no complementizer.[10]

(36) K'iche' extraction from AspP: wi in the embedded clause (M&R 2021:11); embedding bracketed

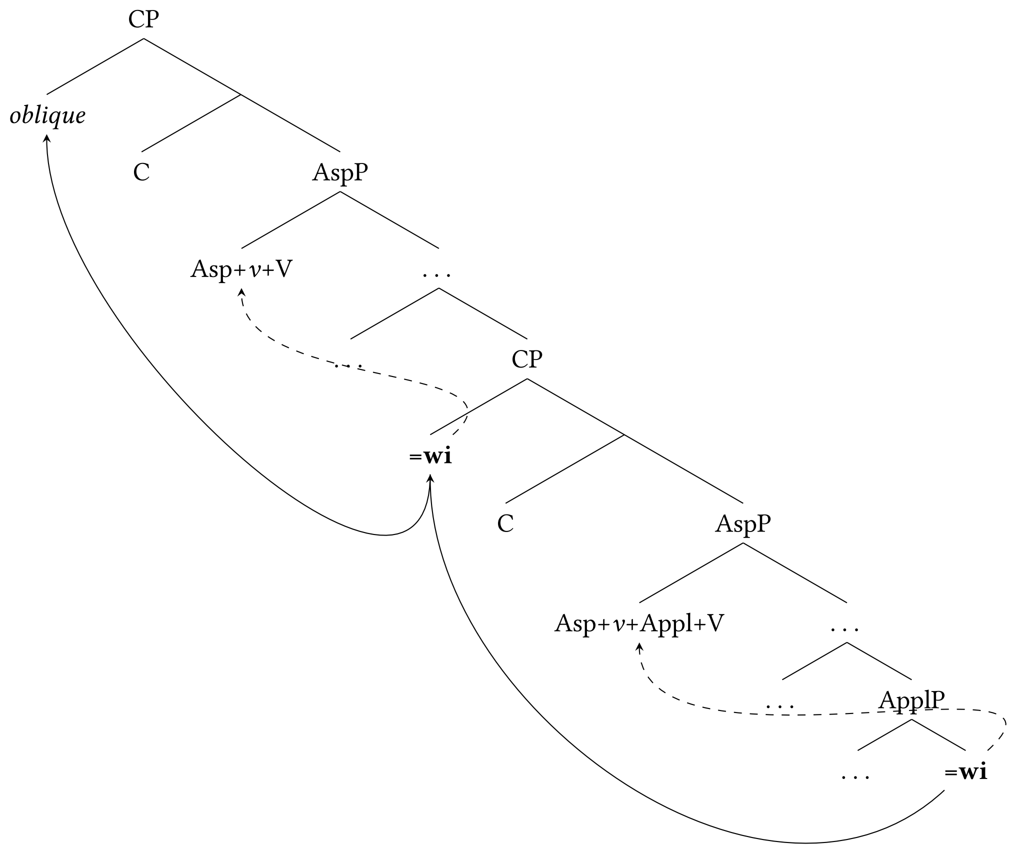
Jas	r-uuk'	k-∅-a-rayii-j	(***wi**)	[	k-∅-a-tij	*(**wi**)	le	wa	]$_{AspP}$?
WH	A3S-RN	INC-B3S-A3S-desire-ACT	**MVMT**		INC-B3S-A2S-eat	**MVMT**	DET	food	

'With what do you desire to eat the food?'

The empirical generalization of the above data is that the presence of *wi* in the matrix clause is contingent on the presence of an overt complementizer in the embedded clause (i.e. the embedded clause must be a full CP; see 11 above). M&R take this generalization to support their claim that *wi* — which they argue to be the spellout of the oblique's movement trace — diagnoses a cyclic phase boundary. That is, when the moved oblique extracts, it must stop over at the edge of the phase (here, the embedded Spec,CP), leaving a copy. This intermediate copy, as well as the lowest copy in the embedded clause, are both spelled out as *wi* (37). In this derivation, M&R additionally propose that the linearization of each pronounced *wi* copy proceeds by uniformly cliticizing upwards onto the preceding predicate: the lower *wi* attaches to the embedded predicate, and the higher *wi* (in embedded Spec,CP) attaches to the matrix predicate (shown in dashed arrows in (37)).

[10] See M&R:7-9 for argumentation in favor of the reduced size of AspP. In what follows, we assume their findings.

(37) Cross-clausal extraction from full CP in K'ichean (M&R ex. (36))



6.2 Long-distance oblique extraction in SJO Mam

We now introduce the empirical facts concerning long-distance oblique extraction in SJO Mam. We first consider full CP-sized embeddings, and then two types of reduced clauses: (i) "aspectless" and (ii) infinitival, which we argue to be of different "sizes", i.e., containing more or less functional structure. In a nutshell, in SJO Mam, we find that if an oblique undergoes long-distance Ā-extraction, the movement enclitic *=(y)a'* may appear in both the matrix and embedded clause, so long as the embedded clause is of a sufficient size (for us, larger than an infinitival embedding). Indeed, we find an asymmetry akin to that found for K'ichean, in that long-distance oblique extraction from (certain types of) reduced clauses limits the possibilities for the realization of the movement enclitic. However, the Mam facts differ in one key respect from those in K'ichean: whereas in K'ichean, the movement particle only appears in the embedded clause in such cases (as in example 36), in Mam, the movement particle may only appear in the *matrix* clause.

First, let us examine full CP embedding in SJO, which involve the following properties: (i) an overt complementizer, such as *kye* or *qa*, (ii) aspect, and (iii) full person agreement on the predicate (shown below in two examples, 38-39). We see in such a case that the extraction of the oblique — here, *se'n=tu'mel* 'how?' — triggers multiple spellout of *=(y)a'*, once on the matrix predicate and once again on the embedded predicate.

(38) SJO extraction from CP: multiple spellout of *=(y)a’* #1
Se’ntu’mel in tma’n(a’) Li’y kye tzaj wulin(a’) Xwan weye’?
se’n=tu’mel in ∅-t-ma-’n**(=a’)** Li’y [kye ∅-∅-tzaj wuli-n**(=a’)**
how=ENCL INC B2/3S-A2/3S-say-DS=**MVMT** María COMP COM-B2/3S-DIR insult-ANTIP=**MVMT**
Xwan w-e=ye’]$_{\text{CP}}$
Juan A1S-RN:DAT=1S
‘How does María say that Juan insulted me?’

(39) SJO extraction from CP: multiple spellout of *=(y)a’* #2
Ja kub’(a’) tximana(ya’) qa jaw(a’) ktx’jo’n(a’) kamisj?
Ja ∅-∅-kub’**(=a’)** t-xima-n=a**(=ya’)** qa [∅-∅-jaw**(=a’)**
where COM-B2/3S-DIR=**MVMT** A2/3S-think-DS=2S=**MVMT** COMP COM-B2/3S-DIR=**MVMT**
k-tx’jo-’n**(=a’)** kamisj]$_{\text{CP}}$?
A2/3P-wash-DS=**MVMT** shirt
‘Where did you think that they washed the shirt?’

Full CP-sized embeddings are also observed in embedded questions (EQ’s): here, instead of a complementizer, the left edge of the embedded clause contains a *wh*-expression, which we take to be in the embedded Spec,CP. Because this *wh*-expression has only moved to the left edge of the embedded clause, and has not undergone long-distance extraction to matrix Spec,CP, we only expect to find MVMT appearing on the embedded predicate, which is indeed the case, as shown in (41).[11] This data will be summarized in the third row in Table 4 below.

(41) SJO embedded question: spellout of *=(y)a’* restricted to embedding
Mixti’ b’i’n wu’ne’ se’n kub’ tbincha’n(a’) Li’y bisiklet.
mixti’ ∅-b’i-’n**(*=a’)** w-u’n=e’ [se’n ∅-∅-kub’ t-b’incha-’n**(=a’)**
NEG B2/3S-know-PASS=**MVMT** A1S-RN:AGT=1S how COM-B2/3S-DIR A2/3S-repair-DS=**MVMT**
Li’y bisiklet]
María bicycle
‘I don’t know how María repaired the bicycle’

Next, we find that in SJO certain reduced embeddings also pattern like full CPs, in the sense that *=(y)a’* is licensed in both embedded and matrix clauses if an oblique is long-distance extracted. We present evidence from two types of embedding both independently argued to be smaller than CP, and show that they have differential behavior: (i) so-called ‘aspectless embedded clauses’; and (ii) infinitival clauses. Whereas multiple spellout of *=(y)a’* is licensed in the former, it is impossible in the latter.

[11] We see a parallel phenomenon in relative clauses in which a *wh*-expression moves within the embedded clause and binds a trace (for more on relative clauses in Mam, see Elkins and Brown 2024).

(40) Relative clause with *ja* ‘where’ in SJO
Xi’ya toj k’wik ja wtan(a’)ye’.
∅-∅-xi’**(*=ya’)**=ya t-oj k’wik [$_{\text{FR}}$ ja$_i$ ∅-∅-wta-n**(=a’)**=ye’ __$_i$]
COM-B2/3S-go=**MVMT**=2S A2/3S-RN:in market where COM-B1S-sleep-ANTIP=**MVMT**=1S
‘You went to the market where I slept.’

First, aspectless embedded clauses (using the terminology from England 2017) are introduced by means of certain complementizers, such as the purposive *u'n* 'in order to, for the purpose of'. These clauses lack AM morphology before the verb (hence 'aspectless'), and are additionally characterized by split ergativity: argument marking within this type of embedding is always Set A, regardless of whether the argument is agent, subject, or object ('super-extended ergativity'; England 2017; Elkins et al. 2025). Following Elkins et al. (2025), we take these clauses to be VoiceP-sized small clauses. In (42) below, we see an aspectless embedding introduced by *u'n*; note that the intransitive subject of the passive verb *k'a'yil* 'buy' here is marked with Set A/ergative, evidencing the case split.

(42) *Chu'xh tu'n tk'a'yit chmek' toj kwik.*
∅-chu'xh t-u'n [**t**-k'a'y-it chmek' t-oj k'wik]
B2/3S-be.difficult A2/3S-RN:PURP A2/3S-sell-PASS turkey A2/3S-RN:in market
'It's difficult to sell turkeys in the market' (lit. '...for turkeys to be sold...')

If long-distance extraction targets an oblique in such an aspectless embedding, we find that *=(y)a'* may appear on both the matrix *and* embedded predicate. A sentence building off (42) above is given in, with *ja=tu'mel* 'where' (43):

(43) SJO extraction from aspectless embedding: multiple spellout of *=(y)a'*
Jatu'mel chu'xh(a) tu'n tk'a'yit(a') chmek'?
ja=tu'mel ∅-chu'x**(=a')** t-u'n [t-k'a'y-it**(=a')** chmeky']
where=ENCL B2/3S-be.difficult=**MVMT** A2/3S-RN:PURP A1P-sell-PASS=**MVMT** turkey
'Where is it difficult to sell turkeys?'

An additional example is provided below in (44), showing that multiple spellout of *=(y)a'* is possible in contexts of long-distance extraction from these reduced aspectless embeddings, regardless of which lexical verb (and which directional) are used.

(44) *Ja tajb'il(a') ku'xun tu'n txu'xin(a')?*
Ja t-ajb'il**(=a')** ku'xun t-u'n [t-xu'xi-n**(=a')**]
where A2/3S-want=**MVMT** young.man A2/3SRN:PURP A2/3S-swim-AP=**MVMT**
'Where does the young man want to swim?'

The second type of reduced embedding we discuss are infinitival clauses: unlike the aspectless embeddings shown above, these clauses are notable for the inability to license any core arguments: the verb is marked with the infinitival suffix *-(V)l*, and complements must be introduced by means of oblique relational noun phrases or else must be used in a generic sense (which we take as a type of noun incorporation; see England 1983:218-219). Following Scott (2023), we take these to be no larger than VP-sized embeddings.

(45) *Ok kq'one' Pabl jqol spik'b'il te Li'y.*
∅-∅-ok k-q'o-'n=e' Pabl [jqo-**l** spik'b'il t-e Li'y]
COM-B2/3S-DIR A2/3P-make-DS=2S Pablo open-**INF** window A2/3P-RN:BEN María
'Y'all made Pablo open windows for María'

If an oblique is Ā-extracted from an infinitival complement clause like this, *=(y)a'* may appear on the matrix predicate, but it is impossible within the embedding. In (46), we see this when *t-e Li'y* 'for María' is *wh*-questioned as *alqe* 'by whom'.

(46) *Alqe ok(a') kq'on(a')ye' Pabl ichil kye tal tx'yan?*

al-qe	∅-∅-ok**(=a')**	k-q'o-n**(=a')**=ye'	Pabl	[	jqo-l(*=**a'**)
who-RN:BEN	COM-B2/3S-DIR=**MVMT**	A2/3P-make-DS=**MVMT**=2P	Pablo		open-INF=MVMT

spik'b'il]
window

'For whom did y'all make Pablo open windows?'

At this point, having looked at infinitives, it is useful to consider an alternative analysis of *=(y)a'*, namely, that it is a pronounced movement copy, à la M&R. If this is on the right track, then all of the instances where *=(y)a'* appears are positions where the oblique has landed at some point in the derivation. The copy/trace of that oblique is then spelled out as *=(y)a'*. From this perspective, the distribution of *=(y)a'* in infinitive clauses is unexpected because these are, by hypothesis, clauses where the oblique originates. Hence, we ought to find traces/copies (i.e., *=(y)a'*) in those clauses, contrary to fact. In addition, if the *=(y)a'* is a copy spellout, the apparent relationship to clause size becomes accidental.[12] We return to alternative analytical possibilities below in Section 7.

To summarize, then, we have shown evidence that reduced clauses are not universally able to license *=(y)a'*: whereas aspectless embeddings do, infinitival embeddings do not. This is the first key difference between the Mam facts and the K'ichean facts: in K'ichean, *all* reduced embeddings described by M&R fail to license multiple spellout of *wi*. The second key difference is the locus at which the movement enclitic is spelled out just in case the embedding is too small: in Mam, it is in the matrix clause only; in K'ichean, it is in the embedded clause only (compare (46) with (36) above).

From these facts, we may conclude for SJO Mam that the appearance of *=(y)a'* on a given predicate hinges on the clause being at least VoiceP sized. Table 4 summarizes the empirical facts seen thus far.

	Matrix clause	**Embedded clause**	**Example sentence**
Long-distance extraction from full CP	✓(+clitic)	✓	(38)
Long-distance extraction from aspectless clause	✓	✓	(43)
Long-distance extraction from infinitival clauses	✓	✗(−clitic)	(46)
One-step extraction from full CP (e.g. EQ's)	✗	✓	(41)

Table 4: Summary of possible spellout locations of *=(y)a'* based on clause type

6.3 Extending the analysis to long-distance movement

In this section we show that the analysis presented in §5 can be extended to the long-distance extraction configurations discussed above. For space concerns, in the derivations below we do not show additional intermediate movement through Spec,DirP, although this is entirely possible.

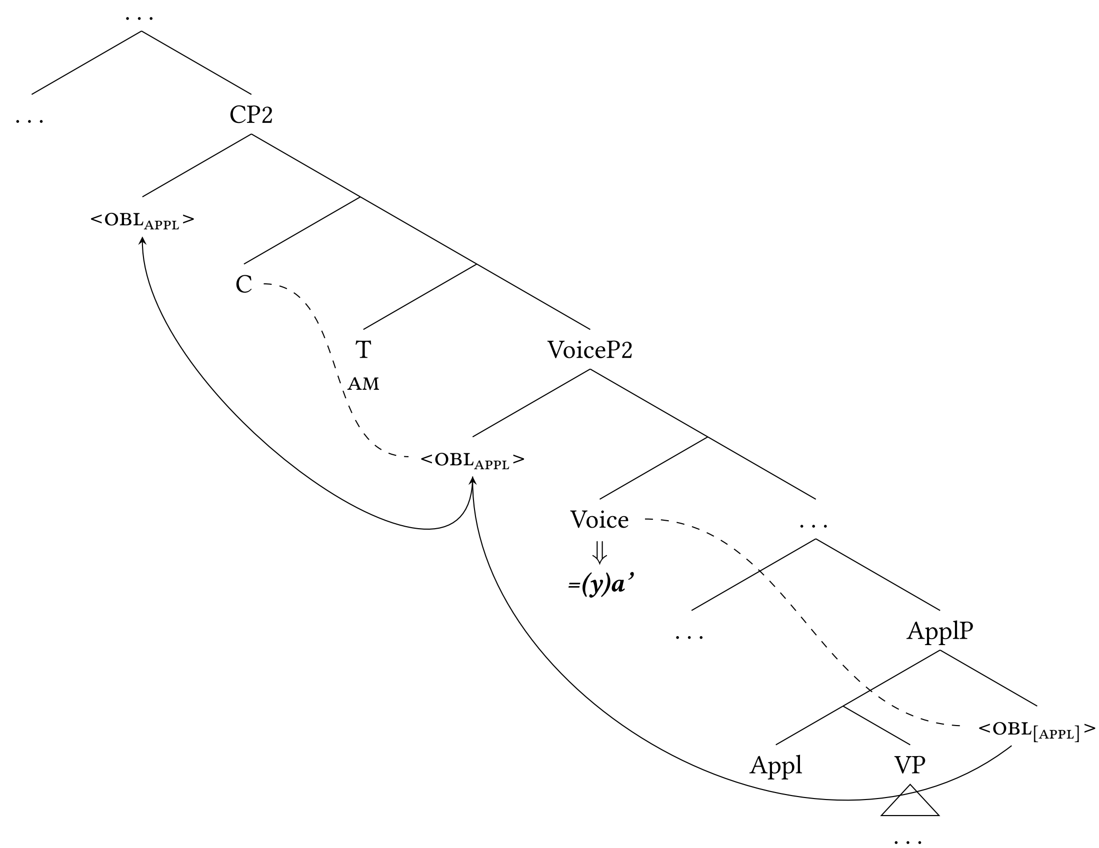
In (47), we schematize oblique extraction from full CP-sized embedding, complete with CP and TP layers. (47a) shows the first steps. Voice^0 first Agrees with the oblique and moves it to its specifier

[12] Regarding finite clauses M&R argue that in long-distance Ā-extraction of the relevant obliques, the oblique first moves to the edge of the embedded CP, and then finally lands in matrix Spec,CP. This successive cyclic movement results in two lower copies of the extracted oblique: one in its base-generated position, and then one in embedded Spec,CP. These lower copies are then spelled out as *wi*, and then both of these morphemes must encliticize leftward onto the nearest predicate (we show this derivation in (37)). For SJO, this derivation would be rather unusual as we do not know of cases in SJO where clitics are able to cross finite clause boundaries.

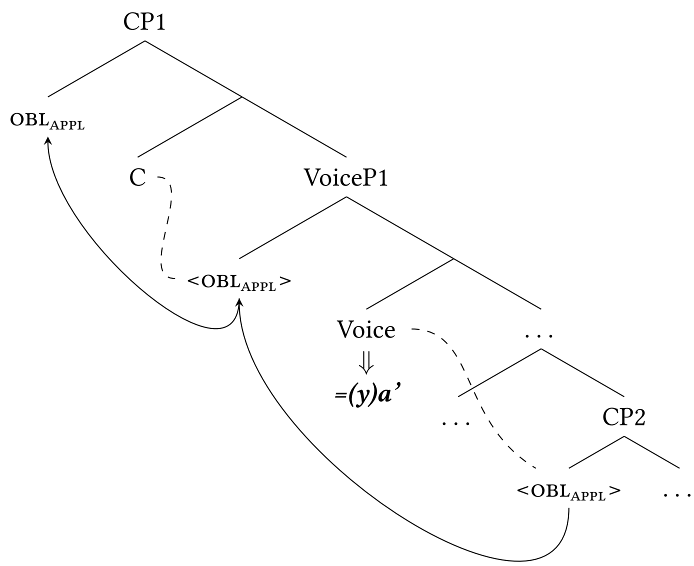
(Spec,VoiceP2), realizing *=(y)a'* as a reflex of this Agree relation. The oblique then continues to the specifier of the embedded CP (labelled 'CP2'). In (47b), the main clause Voice^0 Agrees with the moved oblique in Spec,CP2, moves it to its specifier, and is spelled out as *=(y)a'*. Finally, the oblique moves to Spec,CP1. This derivation accounts for the multiple instances of MVMT along the long-distance movement path.

(47) Full CP embedded clause

a.



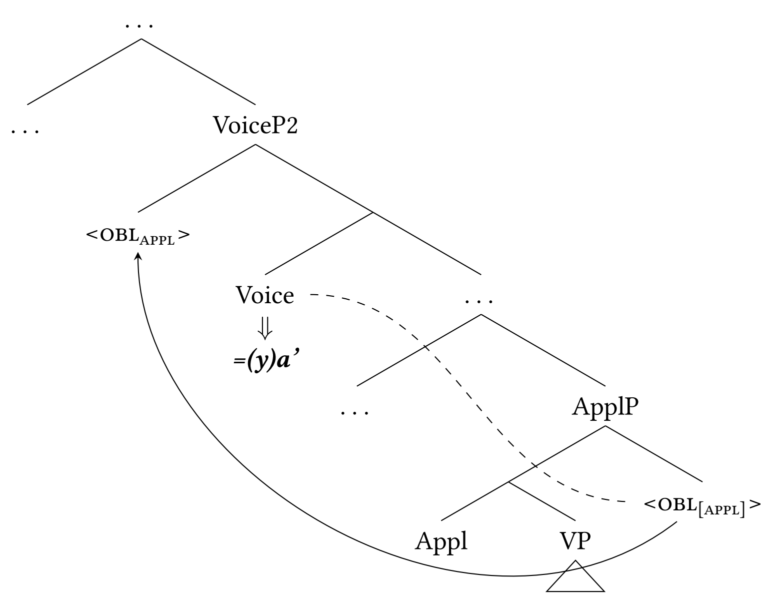
b.



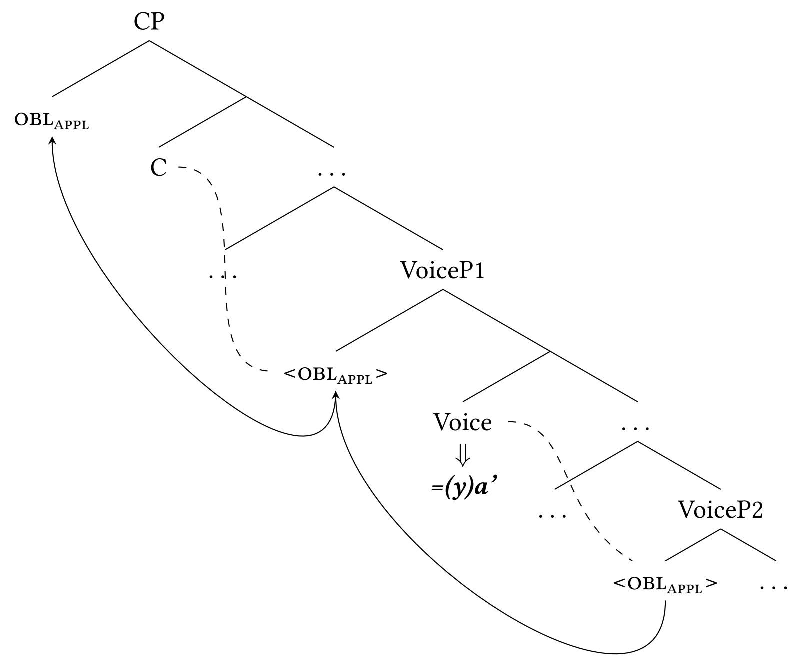
In (48), we schematize oblique extraction from an aspectless embedding, which we treat as a VoiceP-sized small clause lacking a TP and CP projection (Elkins et al., 2025). In (48a), the Voice head of the aspectless embedded clause (VoiceP2) Agrees with the oblique, moves it to Spec,VoiceP2, and is spelled out as *=(y)a'*. In (48b), the oblique continues its movement through matrix Spec,VoiceP, yielding the predicted second occurrence of *=(y)a'* in the matrix clause, and finally lands in Spec,CP.

(48) Aspectless embedded clause

a.

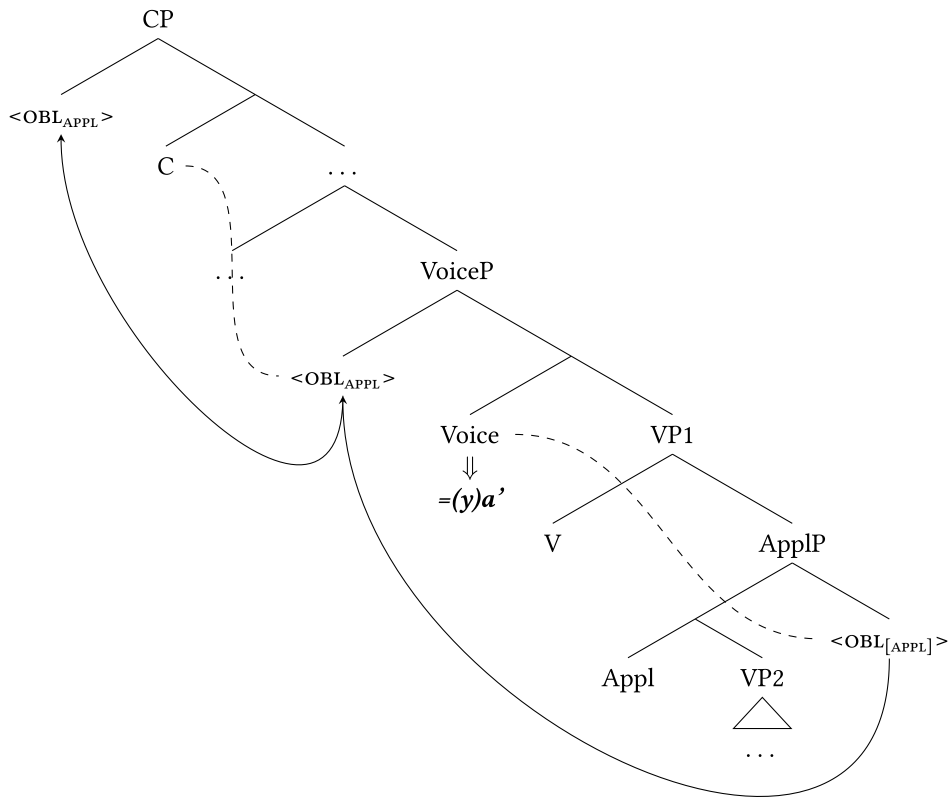


b.



Lastly, we examine oblique extraction from a highly reduced infinitival embedding, which we analyze as containing an embedded ApplP but crucially no VoiceP (Scott, 2023). In the derivation in (49), the matrix Voice head Agrees with the embedded oblique, moves it to its specifier, and is spelled out as *=(y)a'*. The oblique then proceeds to Spec,CP. This reduced embedded structure accounts for the absence of *=(y)a'* in the embedded clause.

(49) Infinitival embedded clause



We note that M&R:16–18 also consider analyzing K'ichean *wi* as the spellout of Ā-agreement, but argue against this on language-internal grounds. They evaluate a hypothetical analysis in which an Ā-head in the CP domain Agrees with the oblique, moves it to its specifier, and spells out *wi*, which then cliticizes downward onto the predicate. For M&R, the problem arises in reduced clauses that lack CP: on such an approach, there would be no way to derive *wi* on the embedded predicate, since the relevant Ā-head would be absent.

SJO Mam does not face this issue under our analysis. Rather than locating the relevant Ā-head in C^0, we identify heads within the verbal domain (namely Voice^0 and Dir^0) as responsible for the spellout of *=(y)a'*. Crucially, when these heads are absent in the embedded structure—specifically in infinitival complements—the movement enclitic *=(y)a'* does not appear on the predicate. We therefore conclude that M&R's rejection of an Ā-agreement analysis is not supported by the Mam facts.

7 Against alternative analyses

Having developed our analysis for monoclausal and long-distance oblique extraction in SJO, we turn now to consider some alternative analytical possibilities that were first raised by M&R in their investigation of K'ichean. We supplement our discussion with additional SJO-specific sources of evidence.

7.1 The movement enclitic is not a resumptive pronoun

The first alternate analytical possibility we consider is to identify *=(y)a'* as a resumptive pronoun, co-indexed with the extracted oblique phrase. Such an approach is also considered, but ultimately rejected, for *wi* in K'ichean by M&R (see M&R:14-15 for their language-specific argumentation). In SJO, we present novel evidence against this approach, specifically that *=(y)a'* may not appear inside extraction islands, which by hypothesis can be ameliorated by resumption (e.g., Ross 1967; Sells 1984; Aoun et al. 2001; Asudeh 2012; Beltrama and Xiang 2016; Ackerman et al. 2018, cf. Heestand et al. 2011).

First, consider the fact that Ā-movement in SJO is subject to island constraints. (50a)-(50b) show a relative clause modifying an object and the formation of a subject *wh*-question, respetively. It is not possible to extract the subject of the relative clause out of it, regardless of whether the antipassive (50c) or active verb forms are used inside the relative clause (50d).

(50) a. *Xi' telq'a'n Xwan chmek' aju tzaj klq'o'n qya toj k'wik.*
∅-∅-xi' t-elq'a-'n Xwan chmek' $[_{\text{RC}}$ aju ∅-∅-tzaj k-lq'o-'n
COMPL-B2/3S-DIR A2/3S-steal-DS Juan turkey REL COMPL-B2/3S-DIR A2/3P-buy-DS
qya t-oj k'wik]
woman A2/3S-RN:in market
'Juan stole the turkey that the women bought at the market'

b. *Alkyeqe' qya lq'on te chmek'?*
alkye=qe' qya lq'o-n t-e chmek'
which=3P woman buy-ANTIP A2/3S-RN:PAT turkey
'Which women bought the turkey?'

c. **Alkyeqe' qya xi' telq'a'n Xwan chmek' aju lq'on toj k'wik?*
*alkye=qe' qya ∅-∅-xi' t-elq'a-'n Xwan chmek' $[_{\text{RC}}$ aju
which=3P woman COMPL-B2/3S-DIR A2/3S-steal-ANTIP Xwan turkey REL
lq'on t-oj k'wik]
COMPL-B2/3S-buy-ANTIP A2/3S-RN:in market
Intended: 'Which women were such that Juan stole the turkey that they bought at the market?'

d. **Alkyeqe' qya xi' telq'a'n Xwan chmek' aju tzaj klq'o'n toj k'wik?*
*alkye=qe' qya ∅-∅-xi' t-elq'a-'n Xwan chmek' $[_{\text{RC}}$ aju ∅-∅-tzaj
which=3P woman COMPL-B2/3S-DIR A2/3S-steal-DS Xwan turkey REL COMPL-B2/3S-DIR
k-lq'o-'n t-oj k'wik]
A2/3P-buy-DS A2/3S-RN:in market
Intended: 'Which women were such that Juan stole the turkey that they bought at the market?'

Next, consider locative extraction, which does involve the movement enclitic (51a). It is possible to only have *=(y)a'* on the matrix predicate, however this can only be construed as being a question about the matrix event, namely the robbery, not the buying of the turkey (51b).

(51) a. *Ja tzaj(a') klq'o'n(a') qya chmek'?*
ja ∅-∅-tzaj(=a') k-lq'o-'n(=a') qya chmek'
where COMPL-B2/3S-DIR=MVMT A2/3P-buy-DS=MVMT woman turkey
'Where did the women buy the turkey?'

b. *Ja xi'(ya') telq'a'n(a) Xwan chmek' aju tzaj klq'o'n qya toj k'wik?*

ja	∅-∅-xi'(=ya')	t-elq'a-'n(=a')	Xwan	chmek'	[RC	aju	∅-∅-tzaj	k-lq'o-'n
where	COMPL-B2/3S-DIR	A2/3S-steal-DS	Juan	turkey		REL	COMPL-B2/3S-DIR	A2/3P-buy-DS

qya	t-oj	k'wik	]
turkey	A2/3S-RN:in	market	

'Where did Juan steal the turkey that the women bought?'

In other words, the oblique cannot be interpreted as though it originated inside of the relative clause. Hence, only the matrix clause reading is possible. We conclude from this that Mam displays island sensitivity for the extraction of both core and non-core arguments. This conclusion is reinforced by the fact that the *wh*-oblique and MVMT cannot be separated by an island boundary (and the number and positions of MVMT inside of the island has no effect on the ungrammaticality, as indicated in 52a). In addition, putting *=(y)a'* both inside and outside of an island still yields an ungrammatical result (52b):

(52) a. **Ja xi' telq'a'n Xwan chmek' aju tzaj(a') klq'o'n(a') qya toj k'wik?*

*ja	∅-∅-xi'	t-elq'a-'n	Xwan	chmek'	[RC	aju	∅-∅-tzaj(=a')	k-lq'o-'n(=a')
where	COMPL-B2/3S-DIR	A2/3S-steal-DS	Juan	turkey		REL	COMPL-B2/3S-DIR	A2/3P-buy-DS

qya	t-oj	k'wik	]
woman	A2/3S-RN:in	market	

Intended: 'Where did Juan buy the chicken that the women bought at the market?'

b. **Ja xi'(ya') telq'a'n(a') Xwan chmek' aju tzaj(a') klq'o'n(a') qya toj k'wik?*

*ja	∅-∅-xi'(=ya')	t-elq'a-'n(=a')	Xwan	chmek'	[RC	aju	∅-∅-tzaj(=a')
where	COMPL-B2/3S-DIR=MVMT	A2/3S-steal-DS=MVMT	Juan	turkey		REL	COMPL-B2/3S-DIR

k-lq'o-'n(=a')	qya	t-oj	k'wik	]
A2/3P-buy-DS	woman	A2/3S-RN:in	market	

Intended: 'Where did Juan buy the chicken that the women bought at the market?'

These data strongly suggests that MVMT is not a resumptive element or pronoun, at least not of the type that can rescue islands (e.g., Sells 1984; Asudeh 2012). We also conclude from these facts that the appearance of MVMT provides further evidence of oblique movement, given its ability to diagnose islands.

7.2 The movement enclitic is not an extraction facilitator

Next, we entertain an additional alternative analysis in which *=(y)a'* is the spellout of an extraction-facilitating voice morpheme, specifically one analogous to *Agent Focus* (AF) voice prevalent in Mayan languages. In many Mayan languages, incluing Mam (see §3.2), the Ā-extraction of transitive subjects/agents (ergatives) is impossible. In order for the thematic agent to extract, the predicate must be marked with a special AF morpheme (see Coon et al. 2014). In Patzún Kaqchikel, for instance, the extraction of the agent *achike* 'who' in (53) requires the predicate be marked with the AF voice morpheme *-o*.

(53) Patzún Kaqchikel (M&R:18)

a.

*Achike	x-∅-u-tej	nu-way?
who	COMPL-B3S-A3S-eat	A1S-tortilla

Intended: 'Who ate my tortilla?'

b. Achike x-∅-tj-**o** nu-way?
who COMPL-B3S-eat-**AF** A1S-tortilla
'Who ate my tortilla?'

M&R comment on the superficial parallel between the AF morpheme *-o* and the MVMT enclitic: both are required to facilitate the Ā-extraction of some element that would otherwise not be able to undergo extraction. However, M&R ultimately reject this proposal because, among other reasons, the movement enclitic may appear alongside other voice morphology, such as the passive. In SJO Mam, we find a similar distribution of elements: *=(y)a'* freely occurs alongside voice morphemes such as the passive *-njtz*. Crucially, voice morphemes such as *-njtz* 'PASS' *are* in complementary distribution with other voice morphemes such as intransitive *-n* and transitive *-'n*:

(54) *Ja'tumel e' k'a'yinjtz(a') muqin tu'n Xwan?*
ja=tu'mel e' ∅-k'a'yi-njtz(**=a'**) muqin t-u'n Xwan
where=ENCL COMPL B2/3S-sell-PASS=**MVMT** tortilla A2/3S-RN:AGT Juan
'Where were the tortillas sold by Juan?'

Secondly, though Mam is sensitive to the ERGATIVE EXTRACTION CONSTRAINT (refer to discussion in §3), it does not have a dedicated AF morpheme. Instead, ergative extraction is facilitated only by means of an antipassive construction. We find that in contexts of long-distance Ā-extraction of an ergative argument, only the embedded clause (for which the extracted argument was the agent) is required to be antipassivized; the matrix clause remains active (55). This indicates that antipassive voice is "contained" within the embedding.

(55) *Alkye tmaya' qa el sun te spik'b'il tu'n xb'uy?*
alkye ∅-∅-t-ma=ya' [$_{CP}$ qa ∅-∅-el su-n t-e spik'b'il
who COM-B2/3S-A2/3S-say=2S COMP COM-B2/3S-DIR clean-ANTIP A2/3S-RN:PAT window
t-u'n xb'uy]
A2/3S-RN:AGT rag
'Who did you say cleaned the window with a rag?'

This behavior differs from long-distance oblique extraction: *=y(a)'* may appear in both matrix *and* embedded clauses, so long as the embedding is large enough (see §6). We take this as additional evidence that SJO *=(y)a'* is not to be identified as a voice morpheme like AF.

8 Summary and discussion

8.1 Comparative empirical and analytical issues

In comparing the K'ichean and SJO data several analytical questions arise. One issue concerns the fact that in both languages, not all obliques condition the appearance of the movement enclitic when extracted. Specifically, temporal obliques ('when') do not trigger MVMT (§4.6). This means that our characterization of SJO MVMT as related to "obliques" is somewhat too broad. At present, we do not have an explanation as to why MVMT does not occur with temporal *wh*-items. One suggestion is that the temporal *wh*-items are unique in not being (sufficiently) nominal, but are instead adverbial, and hence lack (abstract) Case (though see Emonds 1976, 1987; Larson 1985; McCawley 1988; Caponigro and Pearl 2009). However, we find that certain complex temporal phrases like *jni' hor* 'what hour (of the clock)' in example (56), still fail to trigger MVMT despite appearing to be nominal in nature:

(56) *Jni' hor kub' kjone' chib'j?*

jni'	hor	∅-∅-kub'	k-jon(*=**a**)=e'	chib'j?
how.many	hour	COMPL-B2/3S-DIR	A2/3PL-cut-DS=MVMT=2PL	meat

'At what time (lit. what hour of the clock) did y'all cut the meat?'

We leave the precise featural or categorical characterization of the class of MVMT-triggering items as an open issue, but note that this would seem to apply to all languages discussed here.

One interesting point of difference between K'ichean and SJO is the behavior of reason obliques. In K'ichean, these do not trigger the fronting particle *wi*, but in SJO they do trigger *=(y)a'*. At least for SJO, *tiqu'n* 'why' is plausibly analyzed as a PP, composed of *ti* 'what' and *qu'n* 'because', where *qu'n* is a form of *tu'n*, a relational noun in Mam that introduces purposes and reasons (see 26 and 27), as in (57). This is further supported by the existence of other forms roughly corresponding to 'why' (58).

(57) *Jaw xlikpuna qu'n e' jaw chiyon tx'yan toj b'e'*

∅-∅-jaw	xlik'pun=a	qu'n	∅-e'	jaw	chiyo-n	tx'yan
COMPL-B2/3SG-DIR	jump=2SG	because	COM-B2/3PL	DIR	bark-AP	dog

'You jumped because the dogs barked.'

(58) *Alqu'n jaw xlik'pun(a') Xwan?*

al-qu'n	∅-∅-jaw	xlik'pun(**=a'**)	Xwan
who-RN	COMPL-B2/3S-DIR	jump=MVMT	Juan

'Because of whom did Juan jump?'

Thus, the reason adjuncts pattern like PPs in SJO. As we noted, however, this is somewhat unexpected, given that both *reason* and *purposive* 'why' pattern identically, despite having been argued in previous work to be base-generated in different positions in the clause (Collins, 1991; Starke, 2001; Buell, 2011; Shlonsky and Soare, 2011). For example, Starke (2001) argues, based on English, that *reason* 'why' is base-generated in the left periphery while *purposive* (which he calls "motivational") 'why' in the VP region of the clause. As we have shown, in SJO both such *why*'s condition the appearance of MVMT. At the same time, as we have established above, SJO does not seem to have a form that exactly corresponds to 'why'. Instead, it is a PP-based expression, which we used as the basis for suggesting why it occurs with MVMT. At this point it is relevant to point out that in Patzún Kaqchikel, however, 'why' also appears to be PP-like and does not trigger the *wi* marker, as M&R observe:

(59)

Achike	ru-ma	x-Ø-samäj	(*wi)	ri	a	Juan?
what	A3S-RN	COM-B3S-work	(*FP)	DET	CLF	Juan

'Why did Juan work?' (M&R, (14a) [sic])

Thus, it remains unclear why 'why' patterns differently in these languages.

Another point of difference just between SJO on the one hand and K'iche' and Kaqchikel on the other involves adherence to M&R's *Fronting Particle Generaliztion* (given in 11 earlier), which (to paraphrase) states that in single clause embedding cases, the presence of *wi* in the matrix clause is dependent on the presence of an overt complementizer in the embedded clause. Under their analysis, the clauses that lack an overt complementizer are not CPs, but (smaller) AspectPs and that *wi* is the overt spellout of an oblique's trace. They reason that the oblique transits Spec,CP and that the trace is spelled out as *wi*, which then cliticizes onto the matrix clause verb. That is, the source of the *wi* attached in the matrix clause is the embedded Spec,CP. For an Ā-moving oblique originating in a complementizerless embedded clause, there is simply no embedded Spec,CP through which it can transit. Therefore, there is no trace to spell out.

The Fronting Particle Generalization does not extend to SJO. That is, under long-distance Ā-extraction, the presence of MVMT in the matrix clause is independent of the presence of an overt complementizer in the embedded clause (i.e., independent of the size of the embedded clause). The difference in patterns across the languages is related to the proposed mechanisms. In M&R, aside from its base position, the oblique only transits or ends up in a Spec,CP. In SJO, on the other hand, in addition to its base position, an oblique transits multiple positions in a single clause. Indeed, under the analysis proposed in this paper, MVMT is not associated with (Spec,)CP at all: it is a realization of agreement lower down in the clause. In the cases that we have examined, it is difficult to associate MVMT with (Spec,)CP since it occurs in positions that do not seem to be in the left periphery (e.g. following directionals). At the same time, because it can occur multiple times in a single clause, MVMT cannot be associated only with the base position of the oblique. This brings us back to the mechanism of *Chain Reduction via Substitution*, which M&R use to analyze K'iche' and Kaqchikel. We mentioned that simply carrying this analysis over to SJO is initially problematic because it requires cliticization across a finite clause boundary (in the multiclausal case) and we were unaware of this behavior elsewhere in the language. That is, in our system, the movement enclitic is a *simple clitic*: as an Ā-reflex on Voice^0 and Dir^0, its base-generated position and surface position are the same.

Zooming out, given the analysis that we have pursued here, the SJO data strongly suggest that at least a subset of obliques move through at least one (lower) position in a clause on the way to Spec,CP. If this is correct, SJO fits in with a number of other unrelated languages for which (some version of) such movement has been posited, including Avatime (Major and Torrence 2024), Defaka (Bennett et al. 2012), Dinka (van Urk and Richards 2015) and Indonesian (Cole et al. 2008). (But see Keine and Zeijlstra 2025 for recent reanalysis of some of such purported cases that avoids movement through *v*P-level positions.)

8.2 Summary of description and analysis

This paper examined oblique extraction in San Juan Ostuncalco Mam. Unlike the extraction of core arguments (subjects, objects, agents), oblique extraction in this language is marked by the (optional) enclitic *=(y)a'*. Our investigation built on recent work on the related oblique-extraction morpheme *wi* in Kaqchikel and K'iche' (Mendes and Ranero, 2021). We showed, however, that SJO *=(y)a'* differs from K'ichean *wi* in several key respects: (i) which obliques trigger MVMT when extracted; (ii) whether MVMT is licensed in reduced clauses under long-distance extraction; and (iii) the possibility of multiple occurrences within a single clause.

With respect to (i), we showed that *=(y)a'* appears on the predicate when nearly all obliques (except temporals) are extracted. In K'ichean, temporal and reason extraction fail to trigger *wi*, whereas in SJO reason extraction can. Regarding (ii), both languages allow MVMT to appear on matrix and embedded predicates when the clauses are sufficiently large. They diverge, however, under structural reduction: in K'ichean, *wi* surfaces in the embedded clause when the complement is a reduced AspP, while in SJO, *=(y)a'* surfaces only in the matrix clause when the complement is reduced to an infinitival. Finally, for (iii), SJO *=(y)a'* may appear twice within a single clause—on the predicate and on a directional auxiliary, when present.

Given these properties, we developed an Agree-based analysis of oblique extraction in SJO that departs from M&R's account for K'ichean, and argued that the movement enclitic *=(y)a'* in SJO Mam provides clear evidence that successive-cyclic movement operates in both local and long-distance Ā-movement contexts (Chomsky, 1986, 1995, 2001; Rackowski and Richards, 2005; den Dikken, 2009, 2012a,b; van Urk and Richards, 2015).

We have expanded the typology of morphemes that track oblique extraction in the Mayan languages and shown that the substantial variation across subfamilies casts doubt on a "one-size-fits-all" analysis. Future work should more closely examine extraction-tracking morphemes across the K'ichean branch (beyond K'iche' and Kaqchikel) and in the understudied Western Mam dialect area, where they are also attested (e.g. Tacaná Mam; Munson 1984).

Abbreviations

Glosses not following the Leipzig Glossing Conventions are as follows: A – "Set A person marker"; ACT - "Active"; B - "Set B person marker"; DET - "Determiner"; DIR - "Directional"; DS - "Direction suffix"; FP - "Fronting particle (K'ichean)"; INC - "Incompletive aspect"; MVMT - "Movement enclitic (Mam)"; RN - "Relational noun".

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Competing interests

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